



A robust decision-making approach in climate policy design for possible net zero futures

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ABSTRACT

The energy modelling community's analysis of net zero often relies on approaches that hide the extent of uncertainty. Meanwhile the extent of uncertainty involved in the realisation of net zero is proliferating. Conventional consolidative modelling approaches lack of transparency is distorting decision making and policy design around net zero. This contribution uses the UK's Committee on Climate Changes 6th Carbon Budget as a case study. An exploratory, Robust Decision-Making approach is used to highlight the fragility of conventional UK modelling approaches in shaping national climate policy. A new suite of tools, orientation of analysis and mixed approaches are needed to address the extent of complexity, uncertainty and emergence in possible net zero futures. Only then will robust, inclusive and realisable climate policy be designed.

1. Introduction

An essential part of achieving net zero and eventually net negative emissions is the development of a large-scale Greenhouse Gas Removal (GGR) sector (IPCC, 2023). These could both offset the most persistent remaining emissions, and eventually bring the overall concentration of CO₂ in the atmosphere back down to levels required to avoid dangerous climate change. The UK Committee on Climate Change (CCC) has anticipated the need for the generation of between 60 and 100 MtCO₂ per year of negative emissions by 2050 in the UK (Climate Change Committee, 2020). Technical GGRs *i.e.*, those that use chemical means to store carbon in the geosphere include: Bioenergy Carbon Capture and Storage (BECCS) producing various co-products which would remove 22 MtCO₂ by 2035 and 53 MtCO₂ by 2050, Direct Air Capture (DAC)

accounting for 4.5 MtCO₂ with the balance of removals made up of Biochar (BC) and Enhanced Weathering (EW). Land and bio-spheric storage based GGRs *i.e.*, afforestation makes up to 40 MtCO₂ and wood products - 0.44 MtCO₂. Based on this analysis, UK government has set a target of generating at least 5 MtCO₂ of technical GGRs by 2030 and then 23 MtCO₂ by 2035.

This perspectives article contends that the application of conventional modelling approaches used in academia and in government policy design are subject to substantial flaws when applied to the integration of GGR into a net zero economy - not just for the UK but for all nations with net zero ambitions (Workman *et al.*, 2020). Such contexts are subject to uncertainties which are so substantive that they are considered deep,¹ (Lempert *et al.*, 2003). The application of traditional consolidative modelling tools, such as UK TIMES, ESME or indeed Integrated

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¹ Deep Uncertainty is defined as a situation - where analysts do not know, or the parties to a decision cannot agree on, (1) the appropriate conceptual models that describe the relationships among the key driving forces that will shape the long-term future, (2) the probability distributions used to represent uncertainty about key variables and parameters in the mathematical representations of these conceptual models, and/or (3) how to value the desirability of alternative outcomes. In particular, the long-term future may be dominated by factors that are very different from the current drivers and hard to imagine based on today's experiences. Lempert *et al.*, (2003).

Assessment Models more broadly, not only struggle to address the multiplicity of dimensions (Francis, 2018; Pye et al., 2021; Lo Piano et al., 2023) of net zero but also omits risks associated with GGR thus giving a false impression of the feasibility of their use (Workman et al., 2021).

We advocate for the augmentation of the present suite of decision support tools used to design climate change mitigation policy. There is a need to embrace a combination of methods (Gambhir et al., 2019; Morris et al., 2022) that integrate participatory stakeholder engagement and a multi-model approach (Page, 2018) within which an exploratory modelling philosophy is central. This will increase transparency, allowing for the incorporation of values and context driven issues (Devine-Wright, 2022), and open up the decision space through greater clarity regarding trade-offs and ultimately the adoption of robust net zero policy design. These steps are all integral to the design of, and provision of, better-informed net zero policy (Climate Change Committee, 2019). After all, when addressing an existential threat such as climate change the application of as broad an array of decision support tools as the analytical community has at its disposal makes not only for good science but also good governance. It will ensure that the stakeholders integral to the realisation of climate goals, an imperative for the well-being of humankind, are engaged with throughout the process rather than at the end, which is where they are so often relegated.

2. Climate change, greenhouse gas removal, deep uncertainty and decision support analysis

Climate change is happening at a rate that climate scientists are struggling to model (Financial Times, 2021). The complexity and the gaps in the understanding of the sensitivity of the climate system make uncertainty prevalent when analysing its impacts. The socio-political dimensions of addressing emissions reduction, adapting to the risks and transitioning to a net zero economy compound these uncertainties. The need to grow a GGR sector from a standing start to become amongst the largest sectors globally - rivaling energy and agriculture combined - adds yet more uncertainty. GGR are technologically nascent. The policy, regulation, business model design and markets are not established and must be scaled hugely whilst being retrofitted into the economic and cultural fabric of our societies – all within the next three decades. When a system is so complex and uncertainty is so pervasive, the uncertainty is referred to as deep⁴.

The need to address net zero as a complex and deeply uncertain problem is deconstructed in (Workman et al., 2021). The characterisation of the net zero discourse as complex as opposed to complicated is important. Complex systems are characterised by many interacting components whereby aggregated activity is nonlinear, prone to emergent phenomena and can exhibit hierarchical self-organisation. It impacts a number of issues involved in decision support - most salient to this contribution is the application of the appropriate decision support tool(s) (AU4DM Network, 2019a). Ensuring that the most appropriate analytical tools are matched to the extent of uncertainty is fundamentally important as not all decision support tools are able to manage such extensive or high levels of uncertainty (AU4DM Network, 2019b; Cornelis van Dorsser et al., 2020). This is expanded further in the methods section where the application of an exploratory - Robust Decision Making (RDM) approach - is justified to work with the deep uncertainty in net zero.

3. Robust decision-making approach of the UK's Committee on Climate Change's 6th Carbon Budget (CCC's CBP) strategy for GGR in achieving net-zero

This perspectives contribution applies an exploratory framework to the UK's 6th Carbon Budget, by applying a simple RDM approach to explore pathways to meeting UK net zero climate targets. These insights are posited around four salient issues integral to designing better policy

around net zero policy: (1) The need for greater openness in the analytical process; (2) allowing policy decision makers to better understand the impacts of complexity and deep uncertainty on trade-offs; (3) introducing robustness into analysis outcome; and (4) Importantly, its relative simplicity allows transparency of uncertainty, acknowledgement/appraisal and replicability.

This analysis applies the 4-step iterative RDM approach process (Groves and Lempert, 2007; Matrosov et al., 2013; Haasnoot et al., 2013) - by using energy simulation tool called Tranzparent and Nebula² to undertake the RDM component - for detailed explanation of the methods see Appendix A. Supplementary Information in the online version at doi:10.1016/j.envsci.2024.103886, where data is also presented in an accompanying MSeExcel Workbook. Two iterations of the process have been partially applied. First, an initial strategy has been tested against a range of scenarios and its vulnerabilities identified. Second, the identified vulnerabilities are used to develop alternative strategies that address them. These subsequent strategies are tested against the range of scenarios and the most robust strategy is identified for 2050. This approach is schematised in Fig. 1, with further detail given in the methods.

3.1. Phase A: vulnerabilities in the 6th Carbon Budget

To explore the uncertainty surrounding the implementation of GGR measures - for the CCC's proposed 6th Carbon Budget (CCC's CBP) a trade-off analysis was undertaken for the following metrics: (1) UK electricity system cost – electricity generation, H₂ supply and GGR technologies; (2) annual emissions of the system; (3) minimum capacity margin - the remaining available generation capacity when all the demand is met and should stay above several percent of the actual demand load; (4) the Technology Readiness Level (TRL) of the GGR technology; (5) land footprint of the end to end value chain including that used to cultivate the biomass in the case of BECCS - as an equivalent to UK landmass; and (6) water consumption in km³ pa. The first three metrics were generated by Tranzparent and the latter were evaluated by literature review - see Methods section. The results illustrate the wide range of uncertainty on most of these metrics - Fig. 2. Additional and/or alternate metrics could be used to assess trade-offs in each scenario.

The CBP strategy meets the Net Zero emissions target for 2050 in all scenarios. However, none of the scenarios resulted in a minimal capacity margin higher than 4 % - the minimum recommended (Royal Academy of Engineering, 2013). Indeed, the greatest emissions reductions are only achieved at zero or negative capacity margins, which would be unacceptable and as such are represented by greyed-out lines. Negative system capacity margins are driven by high electricity requirements for DACCS (with an uncertainty range of 1.10–56.5 GJ/t_{CO2}). It is also clear that there are considerable energy system cost uncertainties with achieving the emissions reduction. The wide range on the cost of the net-zero electricity supply in the 2050 strategy is primarily linked to high parameter uncertainty for DACCS CAPEX (£74.0/t_{CO2}) - £1840/t_{CO2}) and BECCS-power removal costs (£11.5–308/t_{CO2}). The uncertainty on the land-use metric is very large, and the worst case is almost certainly incompatible with other land uses if biomass were to be sourced within the UK and would also be challenging to source internationally.

This assessment of the uncertainty within parameterised components of the GGR technologies simulated by the CCC to meet UK the 60 MtCO₂ technical net component of UK net zero effectively allows policymakers to better understand the *vulnerability* of potential deployment portfolios. It therefore draws policymakers into explore and ask how the uncertainties can be addressed. Contrast this with the way that the simulations are presented in the 6th Carbon Budget Report - see Fig. 3,

² <https://www.tranzparent.energy/about> and <https://www.foresighttranzitions.co.uk/nebula>

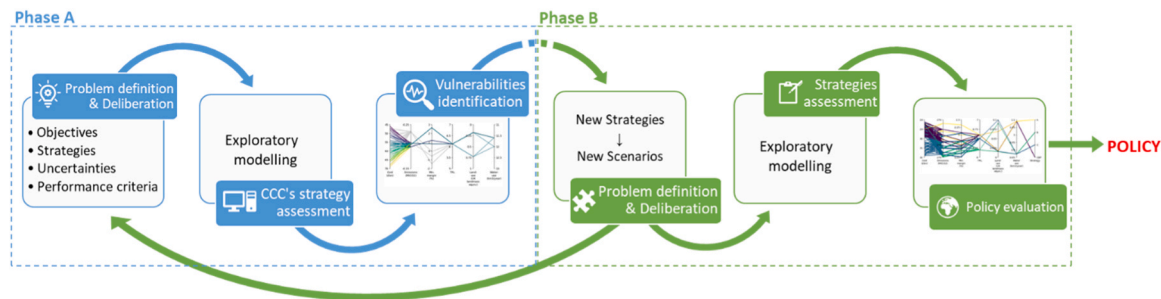


Fig. 1.. A schematic showing how the RDM process was used to establish vulnerabilities in the CCC's GGR strategy (Phase A) then identify alternative strategies and test these (Phase B).

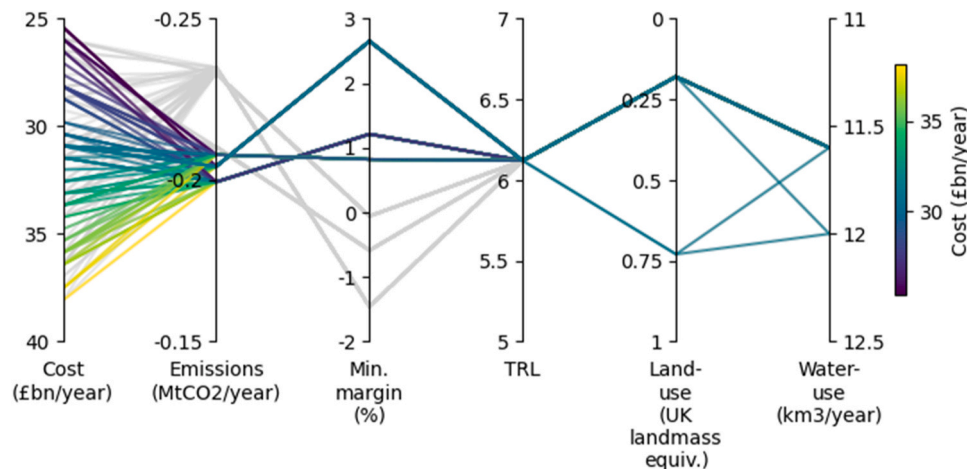


Fig. 2.. Parallel axis plot showing the outcome of the uncertainty analysis involving 144 scenarios with variations in GGR technology parameters on the CCC's Balanced Pathway strategy for GGR in 2050 associated with the realisation of the 6th Carbon Budget. Similar results (not shown) were found for 2030, in relation to the CCC's 5th Carbon Budget. Costs are in £²⁰²⁰.

below.

3.2. Phase B: the performance of alternative strategies to the 6th Carbon Budget: seeking Robustness in GGR Portfolios

To address the vulnerabilities identified in the CBP strategy four alternative GGR strategies were developed - see Table 1 and evaluated against the six metrics - Fig. 4. In strategy A, the smallest capacity of EW and biochar is specified, with EW displacing 3 MtCO₂ of DACCS and biochar 5 MtCO₂ of BECCS-power. This allows a more robust achievement of the zero emissions metric, for a similar minimum generation capacity margin value, a potentially worse land-use metric and a better water use metric. In the base case, shown in Fig. 4a, the overall cost range is similar to that of the CBP strategy; the high uncertainty in DACCS cost is replaced by high uncertainty in EW cost (£10.0–3850/tCO₂). In the analysis shown in Fig. 4b, the range on the EW cost was tightened by discarding the highest and lowest values from the collected estimates giving £44.0–2120/tCO₂. This reduces the cost uncertainty of the strategy from £25.9–40.9 bn^{year}, to £26.0–35.7 bn^{year}, such that it is less than that of the CBP strategy (£25.4–38.1 bn^{year}). Such a reduction in costs might be justified based on the results of field trials reducing the uncertainty range as is taking place in the UK at the moment.³

In the strategies B to D larger capacities of EW and biochar are specified. Strategy B, with the highest deployments of these GGR methods carries significant vulnerability in terms of land use and meeting the net zero target, alongside very high-cost uncertainty with

the un-filtered EW cost range (Fig. 4b), hence is likely not robust.

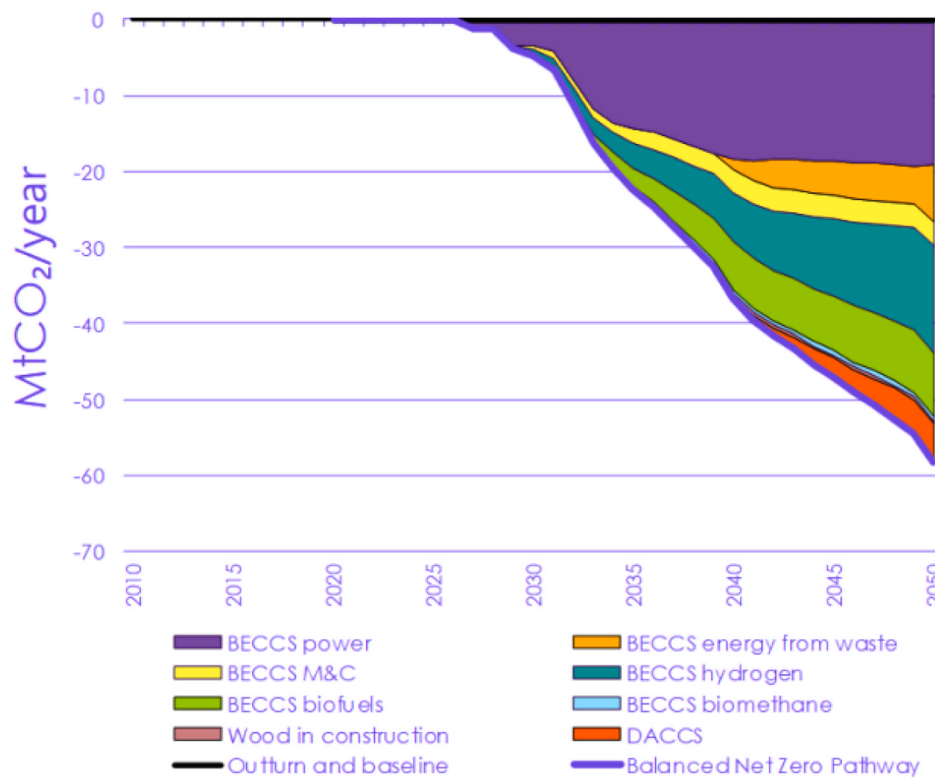
4. Discussion

A relatively basic RDM approach has been conducted to assess the impact of endogenous GGR technology parameter uncertainty (Phase A, 144 scenarios) on the CCC's CBP net zero strategy. Four alternative strategies were then developed and assessed (Phase B, 1280 scenarios). Here, the benefits of the exploratory approach over orthodox consolidative approaches are highlighted.

4.1. Greater transparency - opening up of the uncertainty space

Robustness analysis allows greater transparency in the analytical process by unpacking uncertainty to clearly demonstrate the trade-offs that policymakers must consider. It also circumvents many of the distortive issues around reference scenarios raised by Grant et al (Grant et al., 2020) and avoids the principal agent responsibility issues created by the analytical community conventional data presentation in Fig. 3: *Is it the policymaker that should inquire as to the extent of uncertainty in analysis or is it the analysts that should be more transparent?* A lack of responsibility on both parties part has clearly distorted decarbonisation policy design since the introduction of Integrated Assessment Models over 35 years ago as simulations are taken to be evidence of feasibility (Thompson, 1984). Even with the relatively limited set of strategies presented here, it is clear to policymakers what the strengths and weaknesses of each strategy are and in what circumstances they might fail. Trade-offs are also highly visible, which contributes to better informed stakeholders and will ultimately facilitate adoption of

³ <https://co2re.org/ggr-projects/>



Source: BEIS (2020) Provisional UK greenhouse gas emissions national statistics 2019; CCC analysis. M&C = Manufacturing and Construction.

Fig. 3. The Committee on Climate Changes presentation of the UK GGR portfolio simulated to realise UK net zero by 2050 in the 6th Carbon Budget (CCC's CBP). The figure obfuscates the extent of uncertainty that proliferates the establishment, development and scaling of a new sector as large as the UK water sector in less than three decades. The side-stepping of the complexity involved stands in stark contrast to the projection in Fig. 2 - above - which emphasises the variability in the metrics and accordingly the substantive vulnerability in the CCC's CBP. Additional metrics could be used to assess vulnerability in each scenario.

Table 1

The four GGR strategies introduced in Phase B of the RDM analysis - A to D, alongside the 6th Carbon Budget strategy (Carbon Balanced Pathway, CBP).

Strategy	BECCS-power	BECCS-H ₂	DACCS	EW	Biochar
CBP	30	23	5	-	-
A	25	23	2	3	5
B	23	15	-	10	10
C	25	18	2	3	10
D	25	23	0	5	5

For each GGR technology, the scale target is given in MtCO₂/year.

Key: BECCS - Bioenergy Carbon Capture and Storage - Power and (H₂) Hydrogen | DACCS - Direct Air Carbon Capture and Storage | EW - Enhanced Weathering | Biochar

strategies more likely to be successfully implementable by policymakers. This also potentially facilitates the incorporation of values-based input from a range of stakeholders and audiences in net zero policy design (Workman et al., 2023) - see later.

4.2. Variability in technology parameterisation results in extensive uncertainty

Conventional modelling approaches supporting climate policy design optimise for a fixed set of assumptions. They struggle to provide sufficient relevant insights on uncertainties around system feasibility, actor behaviour, and the effectiveness of interventions which would better inform decision makers when designing policy. The analysis presented in this perspective demonstrates that when considering the

large scale of deployment of GGR technologies required to meet UK climate targets by 2050, variations in techno-economic parameters result in very large uncertainties. An exploratory approach allows these uncertainties to be easily explored, trade-offs to be identified and vulnerabilities in current and alternative strategies discovered - Fig. 4a - and indeed improved as new data becomes available - Fig. 4b. It is noteworthy that the analysis in this perspectives piece is very rudimentary as net zero energy system uncertainty has only focused on GGR. A more systemic analysis of techno-economic uncertainty across the technology portfolio is undoubtedly warranted e.g., the costs of the balance of UK electricity generation system - wind, solar, biomass, gas with carbon capture and storage, hydrogen production etc.

4.3. Attaining robustness in net zero policy design

Traditional decision support tools for climate policy design technology portfolios to realise net zero targets. Such a 'predict then act' mindset tends to lead to fragile strategies which fail to accommodate for deep or disruptive uncertainties such as the six-fold increase in wholesale gas prices in early 2022 due to the Russian invasion of Ukraine which made much of the CCC CBP analysis unfit. Though they returned to long-run levels 18 months later the damage to the credibility of analysis had nonetheless been done (Joffe, 2023). The pervasiveness of complexity and uncertainty in attaining net zero means that robustness becomes the imperative in policy design (Gambhir and Lempert, 2023).

The analysis for this perspective demonstrates that the current CCC CBP strategy has vulnerabilities that make it not robust when compared to alternatives across certain metrics. Assessing vulnerabilities via testing against many future scenarios helps to identify where a strategy

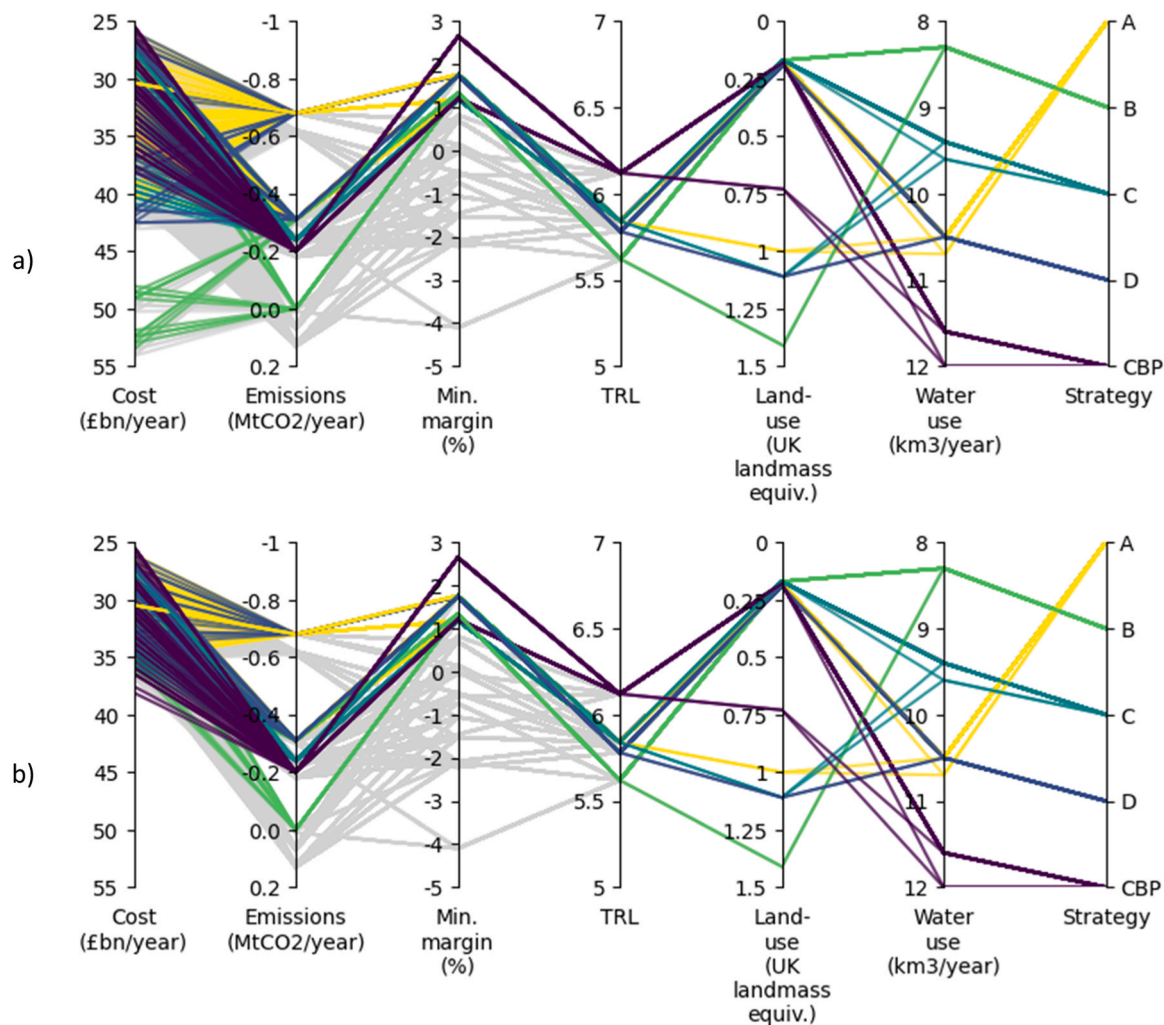


Fig. 4. Parallel axis plot showing the metrics for the full set of 1424 scenarios generated by the analysis for 2050. From left, the first three objectives were quantified using Tranzparent, TRL from a report commissioned by BEIS, and land and water use from a review of existing literature; Sources are given in the methods section. Scenarios that were modelled as having a minimum capacity margin less than 1 %, and/or CO₂ emissions greater than 0 - i.e., do not hit net zero - have been greyed out. In (a), the full range of literature cost estimates for the £/tCO₂ cost of enhanced weathering is applied (£10–3850/tCO₂), whereas in (b) the highest and lowest values were discarded, giving £44–2120/tCO₂.

might fail, where there are compromises, and how strategies could be improved. In this research, considerable cost uncertainties in achieving net-zero via CBP were identified, most strongly influenced by DACCS CAPEX and BECCS-power removal cost uncertainty (£74/tCO₂^{year} - £1840/tCO₂^{year} and £11.5/tCO₂^{year} - £308/tCO₂^{year} respectively).

There is a strong dependency of the selected “most robust strategy” on the performance criteria used. For example, the most robust strategy for 2050 among the CCC’s CBP and alternative strategies is the CCC’s CBP if the energy system cost is used, while strategies A & D, which include BC and EW are most robust if the emission levels relative to net zero are used. This insight is useful when considering how to balance different trade-offs in reaching a preferred policy and therefore where strategies can be more robust across as wide range of metrics as feasible.

5. Conclusions - re-orientation of analysis for more participatory net zero policy design

This perspectives contribution using a small number of strategies with scenario discovery limited to techno-economic uncertainties has identified critical insights – and fundamental vulnerabilities - in current strategies for reaching net zero as developed by the UK’s CCC. It has achieved this by application of an exploratory, robust decision-making

approach, to test strategies under a wide range of future conditions.

The exploratory approach can extend beyond modelling and into the engagement with stakeholders in the analytical process, including the identification of performance criteria and evaluation of trade-offs. In this sense the role of the process is to make the implicit explicit, so that stakeholders and especially policymakers understand the system and decision making is clearly informed. This will mean adoption of policy with a much greater chance of being successfully implemented, an imperative given the scale and speed of the challenge to reach net-zero. Present conventional modelling and analytical approaches to climate policy design tend to be didactic, elite to elite transactions in their development (Beck and Mahony, 2018). Furthermore, the stakeholders who would be able to add substantive nuance and insight to modelling and policy are cut off from the process. This is especially concerning around two dimensions of GGR sector development. Firstly, for GGR sector development, the traditional epistemological process of: academia informing policy which then shapes the market frameworks to incentivise the appropriate positive market actors’ behaviours - does not hold. This traditional epistemological process is not possible in net zero especially around GGR development because the market is running ahead of the ability of policy and academia to underpin the evidence needed. Therefore, the frontier insight and development is actually

being generated by actors within a proto-market. Any process which does not accommodate insight/knowledge generation from present market actors will result in substantive gaps and the potential for miss-aligned recommendations for policy design. Secondly, the need to integrate their value chains through-out the fabric of modern economies and the need for a social dialogue around their acceptability requires a systemic, cross-sector perspective and dialogue with stakeholders and audiences who are not usually engaged in net zero (Helgeson et al., 2022; Anderson et al., 2023). It will avoid a heretical situation whereby calls for broader debate as to the integration of controversial technologies such as GGR into climate policy are made almost two decades after modellers have been integrating them into their tools (Van Vuuren et al., 2017) when the global carbon budget was at least 700 GtCO₂ larger - which likely demonstrates a need for considerations beyond modelling processes (Jasanoff, 2007; Saltelli, 2020).

This cannot happen whilst the majority of funding is appropriated by such a limited suite of tools, inadequate in their ability to address the extent of complexity and deep uncertainty, generating inadequate net zero policy insight (Gambhir et al., 2022; Intergovernmental Panel on Climate Change, 2023). It is hoped that this contribution will stimulate the funding of a more diversified approach - incorporating exploratory and qualitative approaches - to better inform robust climate policy design for possible net zero futures which is transparent and inclusive. Climate change and net zero is after all something that impacts everyone.

CRedit authorship contribution statement

Sol Brown: Writing – review & editing, Supervision. **Lucas Kruitwagen:** Supervision. **Diarmid Roberts:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation. **Mat Wilkes:** Visualization. **Geoff Darch:** Writing – review & editing, Supervision, Project administration, Methodology, Investigation, Conceptualization. **Bastien Denisart:** Writing – review & editing, Writing – original draft, Methodology, Investigation. **Mark Workman:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data supporting the findings of this study are available in the file "Workman et al 2024 - Supplementary materials 2" at: [doi:10.1016/j.envsci.2024.103886](https://doi.org/10.1016/j.envsci.2024.103886)

Appendix A. Supplementary Information

Appendix A associated with this article is "Workman et al 2024 - RDM_ES&P" which can be found in the online version at [doi:10.1016/j.envsci.2024.103886](https://doi.org/10.1016/j.envsci.2024.103886).

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